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PAPER

Isolation and detection of single molecules on paramagnetic beads using sequential fluid flows in microfabricated polymer array assemblies†

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We report a method for isolating individual paramagnetic beads in arrays of femtolitre-sized wells and detecting single enzyme-labeled proteins on these beads using sequential fluid flows in microfabricated polymer array assemblies. Arrays of femtolitre-sized wells were fabricated in cyclic olefin polymer (COP) using injection moulding based on DVD manufacturing. These arrays were bonded to a complementary fluidic structure that was also moulded in COP to create an enclosed device to allow delivery of liquids to the arrays. Enzyme-associated, paramagnetic beads suspended in aqueous solutions of enzyme substrate were delivered fluidically to the array such that one bead per well was loaded by gravity. A fluorocarbon oil was then flowed into the device to remove excess beads from the surface of the array, and to seal and isolate the femtolitre-sized wells containing beads and enzyme substrate. The device was then imaged using standard fluorescence imaging to determine which wells contained single enzyme molecules. The analytical performance of this device as the detector for digital ELISA compared favourably to the standard method, *i.e.*, glass arrays mechanically sealed against a silicone gasket; prostate specific antigen (PSA) could be detected from 0.011 pg mL⁻¹ up to 100 pg mL⁻¹. The use of an enclosed fluidic device to isolate beads in single-molecule arrays offers a multitude of advantages for low-cost manufacturing, ease of automation, and instrument development to enable applications in biomarker validation and medical diagnosis.

Introduction

The ability to isolate and detect single molecules^{1,2} is at the forefront of next-generation analytical techniques in genomics,^{3,4} proteomics,^{5,6} medical diagnosis,^{1,2,7} and chemical and biological warfare detection.^{8,9} We have developed a method based on single-molecule arrays (SiMoA) to isolate and detect single protein molecules that allows for precise measurements at unprecedentedly low concentrations over a broad dynamic range.^{1,2} In this approach, proteins were captured on antibody-coated paramagnetic beads (2.7-μm diameter) and labeled with single enzymes using standard immunocomplex sandwich methods. These beads were then loaded using high centrifugal force into arrays of ~50,000 femtolitre-sized wells that were fabricated from bundles of glass fiber optics, such that wells contained a single bead. The arrays were then mechanically sealed against a silicone gasket in the presence of a fluorogenic enzyme substrate to fluidically isolate the femtolitre-sized wells.

Beads associated with a bound enzyme-labeled immunocomplex or immunocomplexes generated a locally high concentration of fluorescent product confined in the femtolitre-sized wells, enabling the detection of single molecules or singulated ensembles of molecules using standard fluorescence imaging. Other researchers have used similar approaches to isolate molecules in arrays of microwells. For example, Walt *et al.* have used glass arrays sealed against polydimethylsiloxane (PDMS)^{10,11} to study the kinetics¹² and inhibition¹³ of single enzymes in solution. Noji *et al.*^{14,15} used a mechanical plunger to seal arrays of ~300 microwells in PDMS against a planar glass substrate to isolate a variety of enzymes. The use of organic liquids, such as oils, to confine small volumes of aqueous solutions has proven to be a promising strategy across a range of devices, but has mostly been limited to confining picolitre to nanolitre volumes of liquid.^{16–20} Recently, Sakakihara *et al.*¹⁶ reported a method of exchanging aqueous solution with oil on a hydrophilic-in-hydrophobic micropatterned surface to create dome-shaped femtolitre droplets for single-molecule enzyme kinetics studies. Here, we report a method based on polymeric arrays that are sealed with oil in a microfluidic device that greatly simplifies the isolation of single beads in femtolitre volumes, and provides a path toward the development of fully automated single-molecule array systems.

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While our previous approach of sealing glass optical fiber bundle arrays against silicone gaskets has proven reliable to routinely and accurately measure several protein biomarkers^{1,2,21,22,23} and nucleic acids,¹ it presents a number of challenges for developing this technology for use in non-specialized laboratories worldwide. For example, fabricating and assembling femtolitre-sized well arrays using optical fiber glass bundles is expensive, highly manual, and requires a number of precision tools for manufacturing and instrument interfacing. In addition, loading beads in non-planar arrays requires high centrifugal force that is cumbersome and limits the options for automation. Furthermore, the preparation and handling of silicone gaskets to seal the arrays mechanically requires great care, and presents challenges for the robust day-to-day operation. We sought to replace mechanical sealing of glass arrays to silicone gaskets to reduce the cost of the assay and hardware, simplify the isolation and detection of paramagnetic beads, and to enable a highly automated instrument for commercializing the SiMoA technology. We have leveraged tremendous advances in polymer micro-replication technologies, fluidics integration, and imaging technologies to develop a device with these characteristics.

In this report, we will describe the development of enclosed polymeric assemblies that perform the functions of loading and sealing of paramagnetic beads associated with single enzyme molecules in arrays of femtolitre-sized wells using only fluidic flow. We employed a micro-replication technique (injection moulding) using a well-characterized, low fluorescence polymeric material, namely cyclic olefin polymer (COP),²⁴ to fabricate the femtolitre-sized well arrays and the fluidic structures to allow the delivery of fluids to the arrays. Polymer microfabrication technologies have matured over the past decade to enable low-cost and high-volume manufacturing of precise polymer microstructures.^{25,26} The injection moulding technology employed here—which is used to manufacture DVDs—is a prime example of the fidelity (feature size ~300 nm) and manufacture volume (one machine can produce ~10–15 million disks per year) that can be achieved. In addition to the potential cost benefit, micro-fabricated polymer-based array consumables offer the flexibility in array design and the ability to integrate multiple functionalities^{27–29} to facilitate automation. For example, moulding the arrays in a planar format opens up the possibility of flow-based control of bead delivery and removal. Planar formats are also more amenable to processing arrays using automated handling equipment in a high-throughput and reliable fashion. We chose COP for its amenability to manufacture, low cost, favourable chemical and biological properties, and excellent optical properties; cyclic olefin thermoplastics have previously been used in a number of *in vitro* chemical and biological systems.^{30,31} Sealing of the COP arrays was effected using a fluorocarbon oil that was immiscible with the aqueous assay medium. The fluorocarbon oil had the additional benefit of removing excess beads from the surface of the array that can hinder isolation of single molecules. We compared the performance of this device to our current method based on optical fiber bundle arrays and demonstrated comparable results. By integrating bead loading, sealing, and imaging into a single device that is operated with minimal manual intervention, we could enable the development of a fully automated, low-cost instrument for performing single-molecule immunoassays.

Materials and methods

Materials

Optical fiber bundles were purchased from Schott North America. The fiber polisher and polishing consumables were purchased from Allied High Tech Products. Fluorocarbon oil (Krytox[®]) was obtained from Dupont. Non-reinforced gloss silicone sheeting was obtained from Specialty Manufacturing. Hydrochloric acid, anhydrous ethanol, molecular biology-grade Tween-20, and β -galactosidase were all from Sigma-Aldrich. Polystyrene, carboxyl-terminated superparamagnetic beads (LodeStar[™]; 2.7- μ m diameter) were purchased from Agilent. Monoclonal capture antibody to PSA, monoclonal detection antibody to PSA, and purified PSA were purchased from Bio-spacific. Streptavidin, 1-ethyl-3-(3-dimethylaminopropyl) carbodiimide hydrochloride (EDC), and SuperBlock T-20 Blocking Buffer were purchased from Thermo Scientific. Resorufin- β -D-galactopyranoside (RGP), succinimidyl esters of dyes used to label the beads, and magnets used for bead separations in 96-well plates (DynaMag[™]-96 Side Skirted) were obtained from Life Technologies. The anti-PSA detection antibody was biotinylated using the Chromalink[™] reagent (Solulink, Inc.) as described previously;² the average number of biotin groups on the anti-PSA antibody ranged from 7.5 to 9.5. Streptavidin- β -galactosidase (S β G) was conjugated in house using methods described previously;² the average numbers of enzyme and streptavidin molecules per conjugate were 1.2 and 2.7, respectively.

Device fabrication and design

The device was composed of well arrays and fluidic channels that were fabricated in separate planar polymeric parts using an injection moulding technique,^{32–34} and then aligned and bonded to form an integrated polymeric assembly. Fig. 1 shows a schematic for the fabrication process. A positive master for the arrays was first created by laser micromachining of a photoresist-coated silicon wafer. A negative mould was then created from the master by metallic electroforming. A negative mould for the fluidic channels was created by machining steel. COP³⁵ was injection moulded against the two negative moulds; in the case of the fluidic channels, the COP contained 3% carbon black that facilitated laser bonding of the two parts. The two parts were then aligned and laser bonded together. Fig. 2 shows the design of the integrated arrays of femtolitre-sized wells and a complementary fluidic channel. 24 arrays were arranged radially in a disk to enable fabrication using DVD manufacturing processes. Each array consisted of 216,000 femtolitre-sized wells, arranged in a hexagonal close-packed pattern in a rectangular 3-mm $\times$ 4-mm patch on a planar surface. Individual femtolitre-sized wells had nominal dimensions of 4.25- μ m diameter, 3.25- μ m depth, and 8 μ m center-to-center spacing. The fluidic channel was 0.5 mm deep and the total volume of the channel and fluidic inlet port was ~74 μ L to accommodate solutions of beads and sealing oil. The channel contained a constriction to reduce back flows of liquid. The geometry and morphology of the well arrays were characterized using a white light interferometer (Veeco Instruments Inc.) and a scanning electron microscope (SEM) (S-4000, Hitachi or Carl Zeiss SMT). Measurements using SEM were

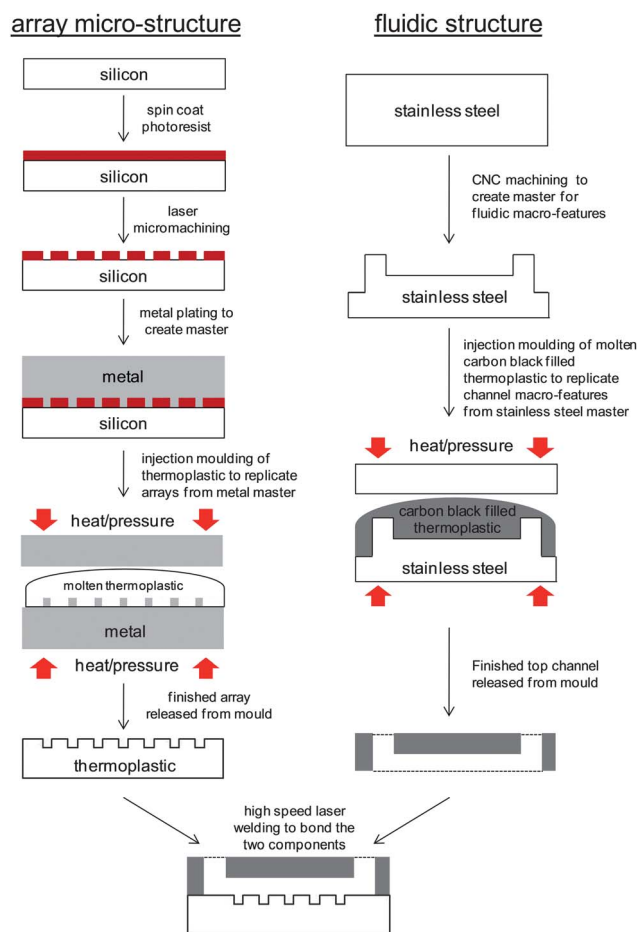


Fig. 1 Schematic of the fabrication process for creating the polymer array assemblies.

calibrated against a USAF-1951 test target (Newport Corporation).

Preparation of antibody-coated and fluorescently-labeled paramagnetic beads

2.7- μm diameter carboxyl-terminated paramagnetic beads were functionalized with a monoclonal antibody to prostate specific antigen (PSA) using EDC coupling according to the manufacturer's instructions. After conjugating antibody onto the bead surface, the paramagnetic beads were functionalized with fluorescent dye, allowing straightforward determination of bead locations in the well arrays using fluorescence microscopy. In this process, 1 mg of Alexa Fluor 488 succinimidyl ester was dissolved in 100 μL DMSO to make a 10 mg mL^{-1} stock solution. 10 million antibody-coated beads were suspended in 600 μL of 0.1 M sodium bicarbonate buffer, pH 8.5. 4.4 μL of the 10 mg mL^{-1} Alexa Fluor 488 succinimidyl ester stock solution was added to the bead suspension. The reaction proceeded for 2 h at room temperature with continuous mixing. The beads were washed five times with PBS and 0.1% Tween-20, with a magnetic separation performed at each wash cycle. Beads were stored in a buffer containing 50 mM Tris, pH 7.8, 150 mM NaCl, 10 mM EDTA,

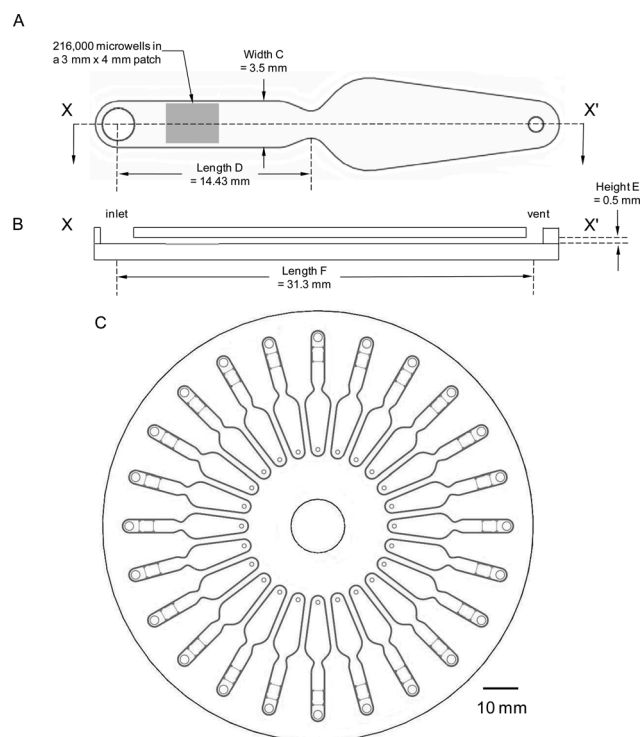


Fig. 2 The design of the polymer array assemblies. (A) Each assembly consists of a fluidic inlet port, a 500- μm -deep fluidic channel that encloses a 3 mm $\times$ 4 mm array of microwells, and a venting port. Around the array, the channel has a width of 3.5 mm; (B) Cross-sectional view of the array assembly along XX' showing the channel height of 500 μm (Height E). The depth of the well array patch is exaggerated to aid visualization; (C) Each device is composed of 24 of these assemblies arranged radially on a 120-mm diam. disk.

1% bovine serum albumin, 1% Triton X-100, and 0.15% ProClin 300. Before and after dye and antibody conjugation, the beads are analyzed using a multisizer (Beckman Coulter Multisizer 4), to ensure that the conjugation and washing steps did not induce bead aggregation.

Capture and enzyme-labeling of prostate specific antigen (PSA) on paramagnetic beads

Test solutions (100 μL) composed of PSA spiked into 25% bovine serum in PBS with 0.1% Tween-20 were incubated with suspensions of 500,000 magnetic beads for 2 h at 23 $^{\circ}\text{C}$. The beads were then separated and washed three times in 5 $\times$ PBS and 0.1% Tween-20. The beads were resuspended and incubated with solutions containing detection antibody (1 nM) in 0.5% bovine serum in PBS with 0.1% Tween-20 for 60 min at 23 $^{\circ}\text{C}$. The beads were then separated and washed three times in 5 $\times$ PBS and 0.1% Tween-20. The beads were incubated with solutions containing S β G (40 pM) in PBS with 0.1% Tween-20 and 1 mM MgCl_2 for 30 min at 23 $^{\circ}\text{C}$, separated and washed seven times in 5 $\times$ PBS and 0.1% Tween-20 and once in PBS. The beads were then resuspended in 15 μL of 100 μM RGP in PBS and transferred to the well arrays for detection.

Loading, sealing, and imaging of enzyme-labeled immunocomplexes captured on paramagnetic beads in a polymeric array integrated with fluidic channels using fluid flow

Fig. 3 shows the process for loading, sealing, and imaging single beads in an integrated fluidic array device. In this process, beads associated with enzyme-labeled PSA molecules suspended in 15 μL of 100 μM RGP were loaded into the assembly through the fluidic inlet port (Fig. 2) using a pipette. Vacuum was then applied to the venting port (Fig. 2) to pull the bead suspension into the channel and over the array (Fig. 3A). The beads were then allowed to settle *via* gravity for 2 min (Fig. 3B). 50 μL of fluorocarbon sealing fluid was injected into the channel to

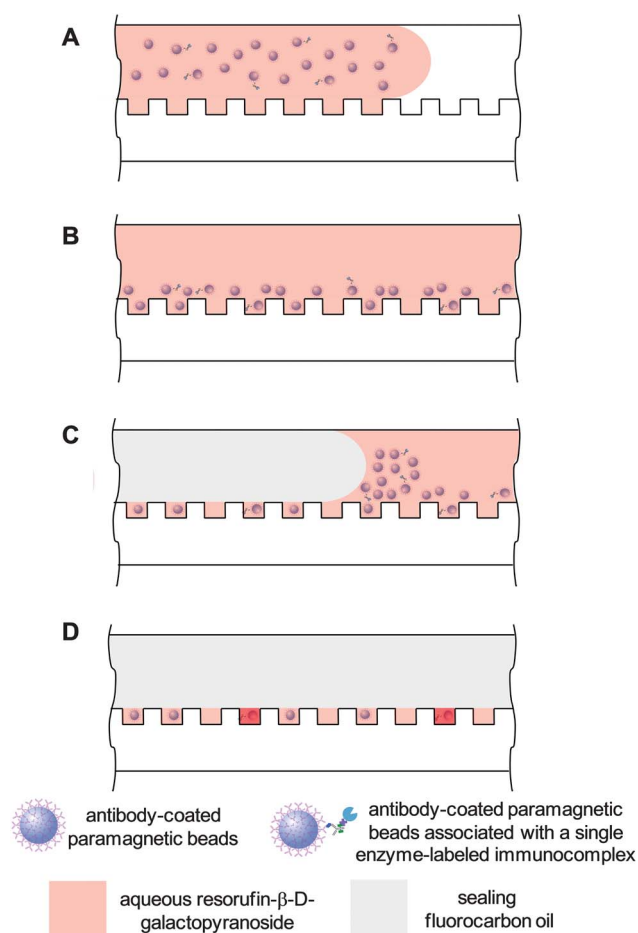


Fig. 3 A schematic diagram of loading, sealing, and imaging of single paramagnetic beads in arrays of femtolitre-sized wells (not drawn to scale). (A) Beads—a fraction of which are associated with captured and enzyme-labeled protein molecules—suspended in 100 μM RGP in PBS are introduced into the array assembly *via* an external pipette (not shown). (B) The beads settle by gravity onto the surface of the array and a fraction of the beads fall into microwells while the remainder are on the surface. (C) Fluorocarbon sealing fluid is introduced into the channel to displace the aqueous medium and excess beads, and to seal the wells. (D) The channel is filled with the fluorocarbon sealing oil and the individual femtolitre-sized wells are sealed and isolated followed by fluorescence imaging. Fluorescent signals are generated in wells that contain beads associated with captured and enzyme-labeled protein molecules. The arrays are imaged through the bottom, clear portion of the device.

displace the bead/RGP suspension, remove excess beads on the surface of the array that did not fall into wells, and to seal the beads in the wells (Fig. 3C). Simulation and flow analysis of this process were performed using Flowworks (Dassault Systèmes SolidWorks Corporation, Concord, MA, USA). The arrays were then imaged through the clear COP on a fluorescence microscope (BX61, Olympus) equipped with a 10.7 megapixel camera (pco.4000, The Cooke Corporation). Five fluorescence images (at 30-sec intervals) were acquired (577 nm excitation; 620 nm emission) with an exposure time of 700 ms using a 10 $\times$ objective to detect enzymatic activity in the wells (Fig. 3D). An additional fluorescence image (495 nm excitation; 521 nm emission) with an exposure time of 1 s was then acquired to locate beads that were labeled with Alexa Fluor 488. The images were then analyzed to determine the fraction of beads that had associated enzymatic activity¹ and to determine the intensity of individual wells.² These data were then converted to average enzyme bound per bead (AEB) as described previously;² the intensity profiles of single molecules in solution were also plotted to determine the kinetics of single enzymes.¹² Arrays fabricated from fiber optical arrays were imaged through their distal end using a custom imager equipped with a 10.7 megapixel camera (Infinity 4–11, Lumenera) as described previously.²

Results and discussion

Device fabrication and characterization

Fig. 4A and 4B show photographs of a representative device that consisted of 24 216,000-well array assemblies in a 120 mm

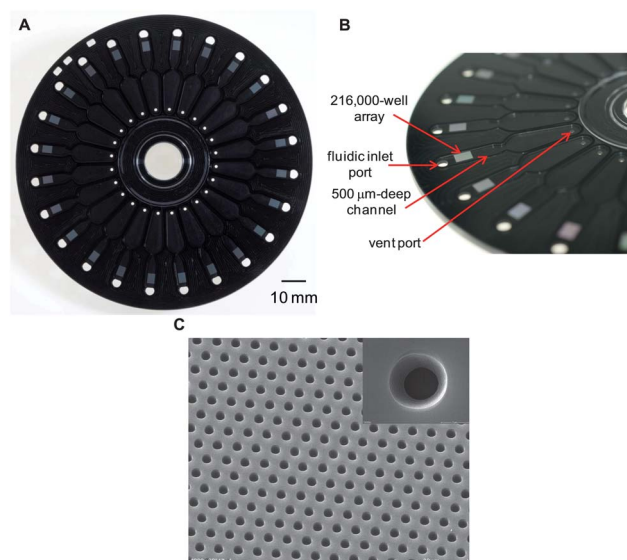


Fig. 4 (A) A photograph of a disk device with 24 array assemblies arranged radially. The array half (closest to the camera) is made of clear COP; the channel half is made of COP impregnated with 3% carbon black, so the disk appears as black. (B) A close-up photograph showing the channel features. The disk was photographed “upside-down” from its functional orientation, *i.e.*, liquids are loaded through the black half and the arrays are imaged from below through the clear half. (C) Scanning electron microscopy images of the injection molded arrays of microwells in COP.

diameter disk format. Accurate measurements of single molecules in these arrays rely on the precise replication of the femtolitre-sized wells. Specifically, when measuring bead-associated enzymatic activity, it is desirable that wells contain either zero or one bead; this constraint requires well-controlled well geometries.¹ Fig. 4C shows an SEM image of the arrays, illustrating the well uniformity and surface flatness that can be achieved by injection molding COP. The widths of the wells were determined by SEM to be $4.222 \pm 0.051 \mu\text{m}$. The depths of the wells were determined using white light interferometry to be $3.274 \pm 0.028 \mu\text{m}$. The results indicated that injection moulding of COP yields structures with acceptable well geometries and dimensional variability. So far we have fabricated and tested hundreds of array disks using the same metallic master without loss of data quality. The use of photolithographically-generated masters enables the rapid iteration of designs and geometries of well arrays.¹ The use of this replication technology also allows arrays with many more wells that translates into greater dynamic range of the single molecule measurements, and opens up the possibility of measuring multiple bead types with the same quality of data.

Integrated polymeric devices to load, seal, and image paramagnetic beads associated with single enzyme molecules

We developed integrated devices composed of arrays of femtolitre wells contained within fluidic structures that would allow the automated delivery of suspensions of beads, enzyme substrate, and sealing oil. These enclosed devices enabled a fluidically-controlled process that would: 1) load beads into the femtolitre-sized wells in the presence of enzyme substrate; 2) remove excess beads from the surface of the array; 3) seal the beads in the wells; and, 4) fluorescently image the arrays of sealed beads. These steps are discussed in detail below.

Loading of beads and substrate. The use of planar arrays integrated with fluidic structures greatly simplifies the delivery and loading of beads into arrays of microwells (shown schematically in Fig. 3A–C) compared to centrifugation of beads in temporary reservoirs as is required for glass arrays.¹ Beads are delivered to the device by a pipette that can be actuated either manually or automatically, enabling integration with robotic liquid handling systems. The high contact angle of the bead solution on COP ($\theta \sim 90^\circ$) meant that the liquid did not enter the channel by capillarity; to draw the liquid into the channel and above the array, a pumping system was used to apply vacuum pressure to the exit of the channel. The device was designed to minimize dead volume and the whole array was covered using $\sim 11 \mu\text{L}$ of bead suspension. The beads—which contain 14% by weight of iron—settle rapidly inside the fluidic channel under the influence of gravity. Fig. 5A and D illustrate a typical outcome of the bead loading process using the fluidic approach inside the array assemblies. Quantitative analyses of images showed that $\sim 45\%$ of wells were occupied by a bead after 2 min of bead settling *via* gravity using 500,000 beads in $15 \mu\text{L}$ (33,333 beads/ μL). This loading efficiency is similar to that achieved loading beads into glass arrays using centrifugation for 10 min at 1,300g, where $\sim 40\text{--}50\%$ of wells contained a bead at bead concentrations of 200,000 per $10 \mu\text{L}$ (20,000 beads/ μL).¹ Bead loading in these

devices is sensitive to bead concentrations and the time allowed for beads to settle; Figure S1 shows bead loading efficiency as a function of bead concentration and time. If all 216,000 wells in the array are imaged, and assuming that at least 100 positive beads need to be imaged to yield an acceptable Poisson noise ($\leq 10\%$) at the typical backgrounds of the digital ELISA (fraction of on beads, $f_{\text{on}} = 0.01$), then a minimum bead loading efficiency of 4.6% is required. Figure S1 indicates that a minimum of 4,000 to 7,000 beads/ μL are required to achieve this bead loading in 2 min.

While bead loading efficiencies of 50% are sufficient for SiMoA experiments, it is desirable to further reduce the time taken to achieve such a bead loading to improve the throughput of instrumentation. Magnets can be used to speed up the delivery of the superparamagnetic beads to the wells. While permanent magnetic fields increase the flux of beads to the wells, they can also cause chaining of beads along magnetic field lines that can reduce the chance of single beads entering single wells. We are currently investigating improvements in magnet designs that reduce chaining but still result in a greater flux of beads to the surface.

Another important aspect of loading, in addition to bead loading efficiency, is consistent filling of the wells with RPG solution. RPG has a relatively high contact angle on COP (90°), and if the microwells had perfectly squared off walls (*i.e.*, a discontinuity of 90° in surface topography), then we might expect that pinning of aqueous solutions at the well entrance would result in poor liquid filling. The features in the moulds used for fabricating arrays, however, had a draft angle of $\sim 7\text{--}10^\circ$ to avoid a perfectly vertical well wall; this draft angle is apparent from SEM images of individual wells (inset of Fig. 4C). This well topography makes it possible for liquid to fill the wells in a reasonable time. We have observed that all wells fill with solutions of beads suspended in RPG within 45 s without any special surface treatments.

Removal of excess beads. Subsequent to bead loading, delivery of an immiscible liquid allowed excess beads to be removed from the surface of the array without affecting those beads in wells. After delivery of beads into the fluidic channels, not all of the beads fall into a well, and, as a result, excess beads remained on the surface of the array. These beads could, potentially, interfere with sealing and also cause extraneous fluorescent signals if not removed from the array area being imaged. For example, when we applied 500,000 beads in $15 \mu\text{L}$ to 216,000 wells during the loading process that resulted in about 45% fill, $\sim 97,000$ beads were loaded into the wells of the array, and about 400,000 remained on the surface. Fig. 5A³⁶ is an optical microscopy image taken after bead loading that shows that a significant number of beads remain on the surface of the array. When using optical fiber bundle arrays, we employed a manual wiping method to remove excess beads. Here, fluid flow of the sealing oil was used to remove excess beads. In this process, the fluorocarbon oil displaces the aqueous medium in the direction of flow, washing away with it the excess beads on the surface, illustrated by the “clean” surface in the trailing side of the sealing front and the aggregation of excess beads along the interface in the direction of flow (Fig. 5B). This method resulted in complete removal of excess beads from the surface (Fig. 5C). Fig. 5D shows an

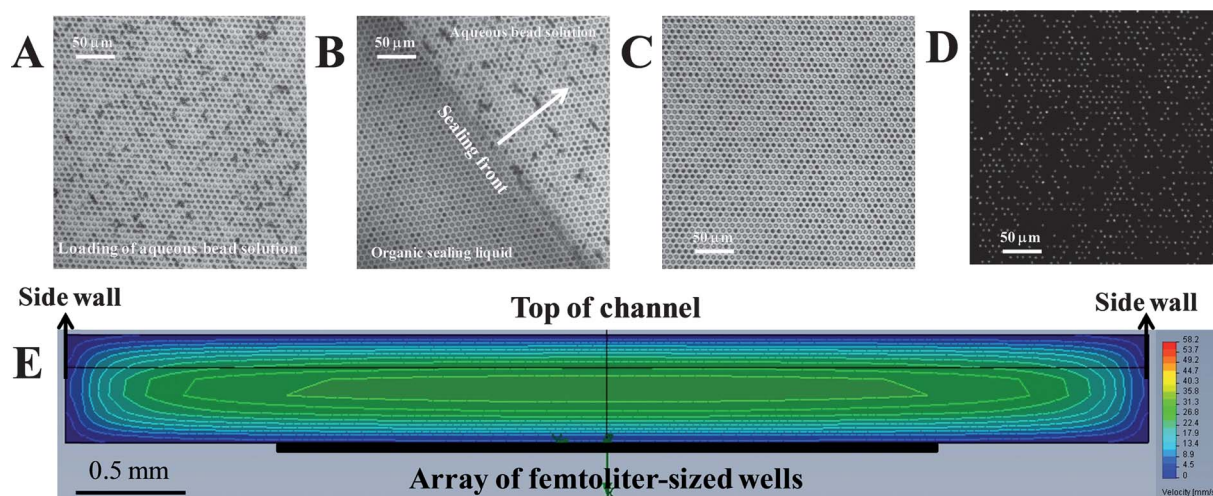


Fig. 5 (A) An optical microscopy image that shows the assay beads being loaded into the femtolitre-sized wells with excess, unloaded beads on the surface. (B) An optical microscopy image that shows the aqueous/oil interface where the excess beads on the surface were being pushed towards the aqueous phase. (C and D) An optical microscopy image (C) and a fluorescence image (D) showing the removal of excess beads upon flowing the fluorocarbon oil while retaining beads that were loaded into the femtolitre-sized wells. (E) A vertical cross-sectional view of a simulated flow profile showing laminar flow of the fluorocarbon oil down the axial direction inside the channel. These images were generated using initial devices that were fabricated using an embossing technique;³⁶ the disks manufactured using injection molding have similar performance.

image of the fluorescent beads on the same part of the array after the oil sealed the array. This image further verifies that the beads that were loaded in the wells remain secured inside the wells after “wiping” by the fluorocarbon oil, suggesting that the net force experienced by the beads is only in the direction of flow. We hypothesized that this phenomenon resulted from the lack of a vertical motion of fluid during flow that could displace the loaded beads inside the wells. To test this hypothesis, we performed fluidic flow analysis using a simulation software package to visualize the flow profile. Fig. 5E illustrates the vertical cross-sectional view of the simulation result showing the flow profile down the axial direction. The laminar flow profile verified that there was no vertical fluid exchange (or mixing) across the aqueous/oil interface as the fluorocarbon oil flows and seals the aqueous content inside the wells. Fig. 5E also indicates that the zero velocity boundary condition creates the highest shear at the array surface, explaining the effective removal of excess surface beads that are held *via* relatively weak gravitational force.

Sealing and imaging of arrays. The critical step in the SiMoA process is the sealing of aqueous solutions in femtolitre-sized wells. Sealing confines the fluorescent product generated from the reaction between an enzyme and its substrate to inside individual femtolitre-sized wells, creating a locally high concentration of product, making single-molecule detection straightforward using conventional optics and cameras used in fluorescence microscopes.¹⁰ For example, when a single enzyme molecule associated with a bead is confined to a single well, more than 28,000 fluorophores are generated on average and localized in <50 fL over the course of 2.5-min data acquisition period, resulting in a local concentration ($\sim 1 \mu\text{M}$) that is easily detected using standard fluorescence microscopes and uncooled CCD cameras. Other approaches to overcome the dilutive effects of diffusion have been developed, *e.g.*, the use of deep, unsealed wells,³⁷ or encapsulating molecules in droplets formed

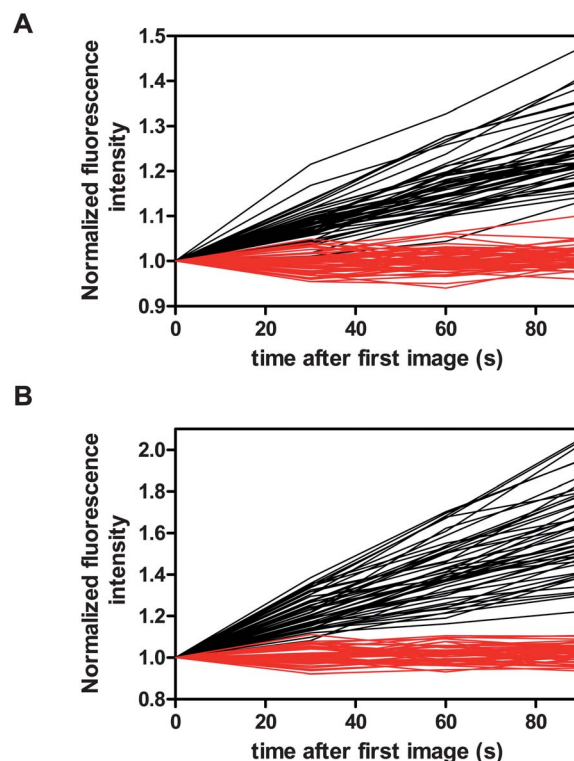


Fig. 6 Measurements of the fluorescence intensity as a function of time relative to the first image acquired from 50 representative “on” wells (black lines) and 50 representative “off” wells (red lines) determined from images of arrays of microwells exposed to solutions of SβG, where the microwells were: (A) optical fiber bundles sealed mechanically against a silicone gasket; (B) polymeric array assemblies sealed with a fluorocarbon oil. The fraction of on wells was <10% in both cases, such that the Poisson distribution indicated that >95% of the “on” wells contained single enzymes.

in multiphase microfluidic systems.^{17,18} Our objective was to develop a simple, robust, non-mechanical method for sealing relatively shallow arrays of microwells so that large numbers of singulated beads could be interrogated. We investigated the use of a fluorocarbon oil (immiscible with the aqueous assay medium) to seal and isolate individual femtolitre-sized wells, as an alternative to using a silicone gasket and a mechanical system. Our premise was that the hydrophobic fluorocarbon oil would preferentially wet the surface of the polymer arrays, thereby displacing the aqueous liquid that the beads were loaded in. The aqueous liquid (*i.e.*, enzyme substrate) within the wells, however, would be in a thermodynamic local minimum caused by surface tension pinning by the change in topography created by the wells,³⁸ allowing the isolation of beads in enzyme substrate in the femtolitre-sized wells. Some of the potential challenges to this approach were: diffusion of fluorescent enzymatic product into the oil phase; displacement of droplets from the wells by the oil over time; and the optical and biological compatibility of the oil.

The kinetics of single enzymes provide a rigorous test of sealing arrays of femtolitre wells.¹² These measurements allowed us to determine the ability to isolate aqueous solution in the wells

using oil, check the stability of the oil-confined solution in the wells, and characterize any possible optical effects introduced by imaging through COP. These data were compared to those obtained from the standard method, *i.e.*, mechanical sealing of glass arrays. Any confounding optical, biological, or stability issue will be apparent from the velocity profiles of hundreds of enzymes within an array. Previous studies have also established these kinetics for comparison,¹² and we used the same experimental set-up from those papers as a control in our experiments. In these experiments, the choice of oil was optimized for its viscosity, to make the droplet of aqueous solution in the femtolitre-sized wells stable for hours, and to provide favourable flow qualities. Fig. 6 shows the kinetics of single enzymes ($f_{\text{on}} < 0.1$) presented as fluorescence intensity change relative to the first image acquired against time for individual wells. These data were generated by isolating solutions of enzyme in arrays using: A) optical fiber bundle arrays sealed mechanically against a silicone gasket;¹⁰ and B) integrated array fluidic devices fabricated in COP and sealed using fluorocarbon oil. The average initial slopes of the fluorescent signals from wells that were identified as being “on”¹ were 0.0034 and 0.0063 normalized arbitrary units/s for the glass arrays and polymeric arrays ($n > 1300$), respectively.

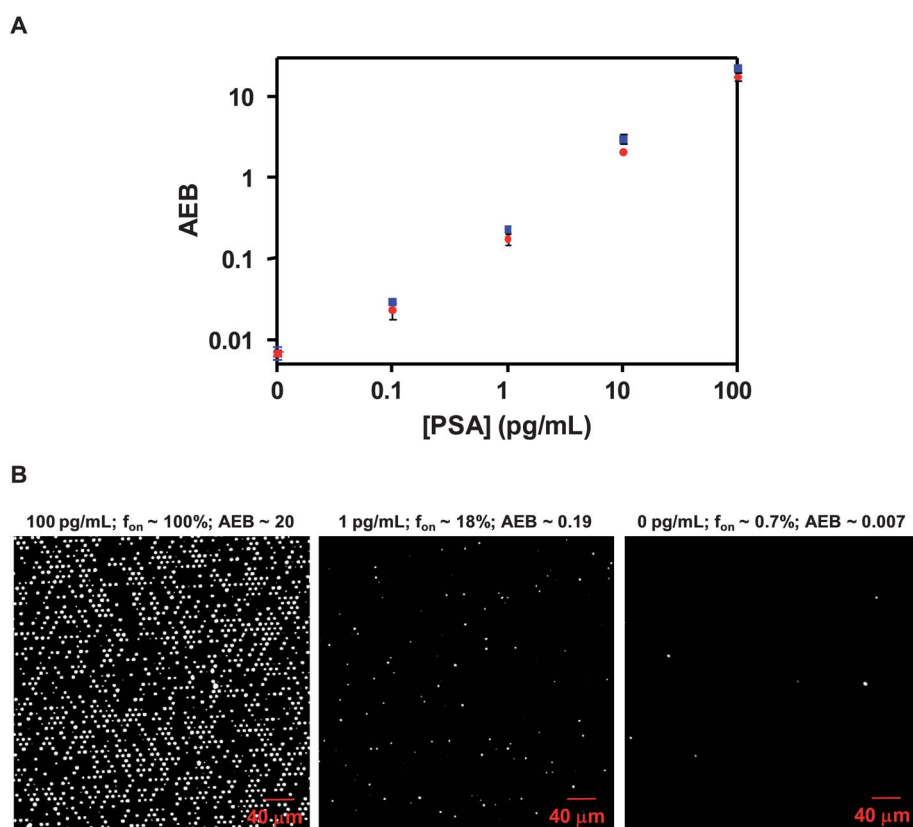


Fig. 7 (A) Plots of average enzymes per bead (AEB) as a function of PSA concentration for digital ELISAs measured using polymeric array assemblies and the method outlined in Fig. 3. The left most point represents the background ([PSA] = 0 pg mL⁻¹) samples. The calculated LODs for two bead lots were 0.011 pg mL⁻¹ (red circles) and 0.016 pg mL⁻¹ (blue squares) based on extrapolating the concentration at the background signal plus 3 s.d. of the background signal. The AEB values at the high end of the analytical range (100 pg mL⁻¹) were 17.6 and 22.2. Error bars are shown for data points based on duplicate measurements; in some cases the s.d. was smaller than the size of the data symbol and the error bars are not visible. (B) Representative images showing the fluorescence signals generated by enzyme molecules associated with PSA molecules on singulated beads at PSA concentrations corresponding to 100 pg mL⁻¹, 1 pg mL⁻¹, and 0 pg mL⁻¹. These images were acquired by performing a fully automated, fluidically controlled load, seal and image of beads in enclosed polymeric array devices.

Differences in absolute slopes between the two methods were due to several differences between the measurements in glass and polymeric arrays, *e.g.*, well volume (~ 46 fL in glass arrays and ~ 38 fL in polymeric arrays), lamp intensity, camera type, imaging through plastic *vs.* total internal reflection imaging through optical glass arrays, and differences in exposure times. It is clear from these data, however, that it is straightforward to identify single enzymes using oil sealing as it is for glass arrays. More importantly in terms of the observed distributions of enzyme turnover rates, the coefficients of variation (CV) in the initial slopes of fluorescence signals from single enzymes were 32% and 32% for glass and polymeric arrays, respectively. The variation in single enzyme activity that has previously been reported in glass arrays was $\sim 30\%$.¹² We conclude that the variation in single enzyme kinetics for two methods is the same and consistent with previous reports. Fig. 6 also shows the tight distributions of “off” wells that did not contain an enzyme, suggesting that the wells were effectively sealed for both methods. These data show that the aqueous/oil interface provides an effective diffusive barrier as achieved using the mechanical seal with a silicone gasket. We conclude that the fluid seal approach is a viable alternative to mechanical sealing that enables robust isolation and quantitation.

Performance of digital ELISA using integrated polymeric devices for SiMoA detection

To test the overall performance of the load, seal, and image approach, we used the enclosed polymeric structures devices for SiMoA detection of a digital ELISA for PSA. We have published extensively on a digital ELISA for PSA using glass arrays and mechanical sealing,^{1,2,22} so it provides a convenient comparison for the performance of the integrated fluidic device. Fig. 7 shows plots of AEB against [PSA] generated using the method shown in Fig. 3 for PSA concentrations ranging from 0 to 100 pg mL⁻¹, for two preparations of anti-PSA beads. The absolute values and slopes of AEB as a function of PSA concentration are similar to those reported previously.² The limits of detection (LOD) in these experiments were determined by extrapolating from a point-to-point fit the concentration at which the signal was equal to the background signal plus 3 s.d. of the background signal. The calculated LODs were 0.011 pg mL⁻¹ and 0.016 pg mL⁻¹, with CVs on the background of 8.4% and 17.7% ($n = 2$), respectively. These values are consistent with the previously reported LOD values that ranged from 0.006 to 0.028 pg mL⁻¹, over several lots of capture beads and biotinylated detection antibodies, and background CV values.^{1,2,22} These devices also performed well at the high end of the dynamic range where average bead intensities (rather than digital counting of active beads) are used to quantify AEB.² At 100 pg mL⁻¹, the AEB values were 17.6 and 22.2, within the values reported previously (AEB $\approx$ 15 to 27) depending on particular reagent lots used.^{2,22} These data, when combined with the enzyme kinetic data (Fig. 6), validate the integrated fluidic approach to measuring single molecules in digital ELISA. Fig. 7 also shows single-molecule fluorescence images of singulated beads associated with enzyme-labeled PSA molecules at three PSA concentrations across the dynamic range of the assay.

Conclusions

We have developed an approach for performing single-molecule isolation and detection in large arrays of microwells in a low-cost polymeric device using fluidic flow. This approach leverages established high-volume polymer replication to realize low-cost, disposable single molecule arrays. These devices greatly simplify process development and automation, and could enable the development of fully automated instrumentation for performing high-throughput testing of samples in single-molecule arrays for applications in life science research and *in vitro* diagnostics.

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